

Genus-0 Virtual Braids



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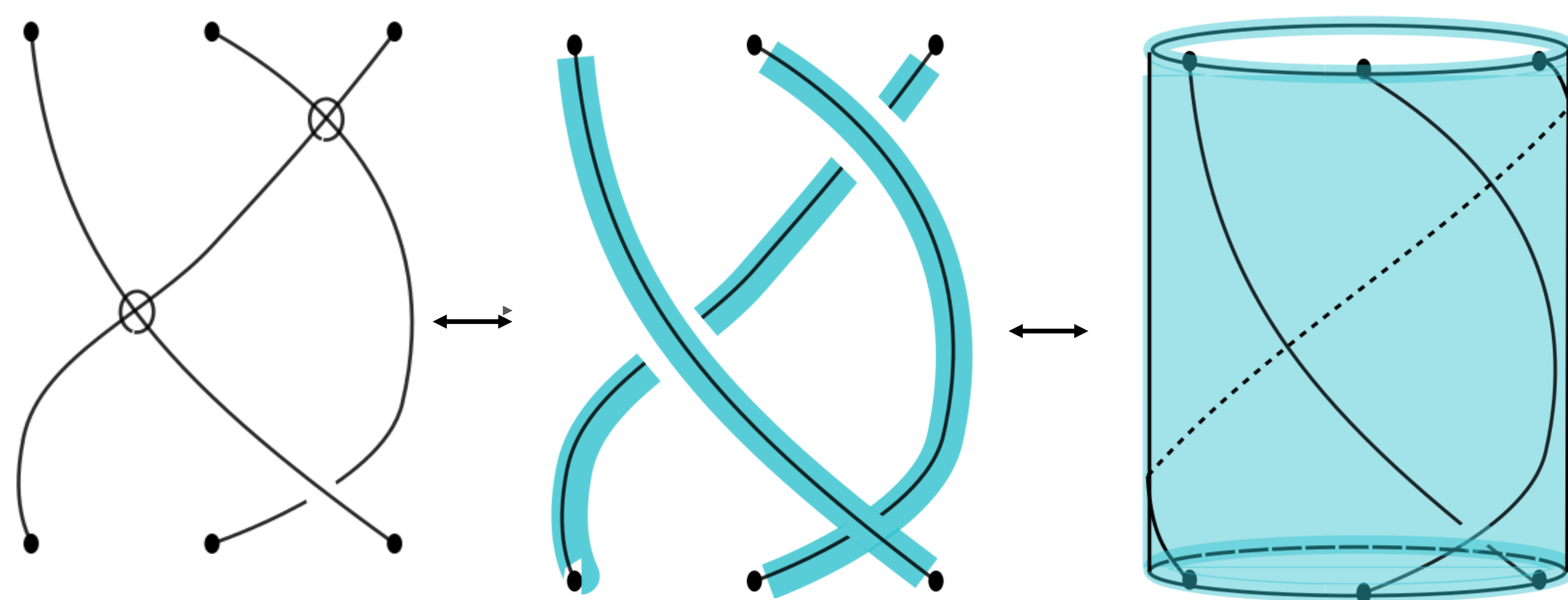
1: Vanderbilt University 2: Georgia Tech

Overview

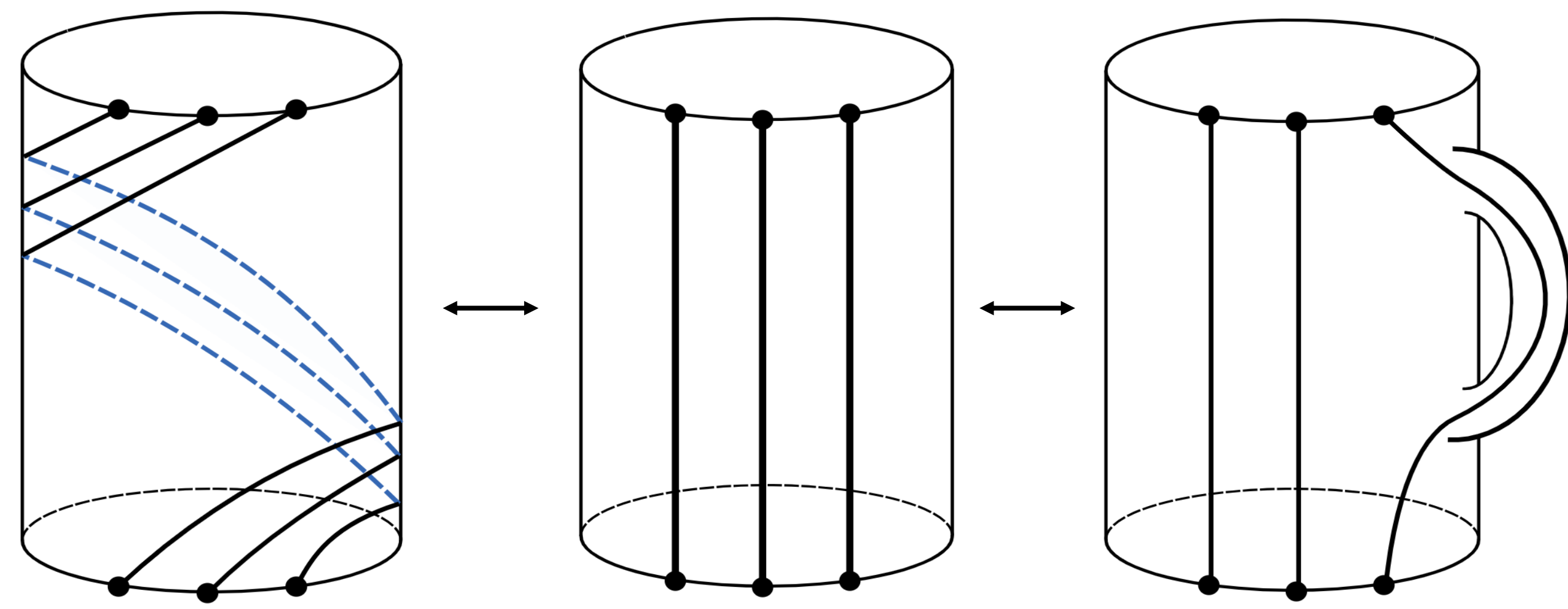
The virtual braid group extends the classical braid group by the symmetric group. We define a new subgroup of virtual braids using a topological property, and we investigate its algebraic structure.

Virtual Braids

There is a bijection between virtual braids and **thickened surface diagrams**.



Thickened surface diagrams are equivalent up to **homeomorphism** and **stability** moves.



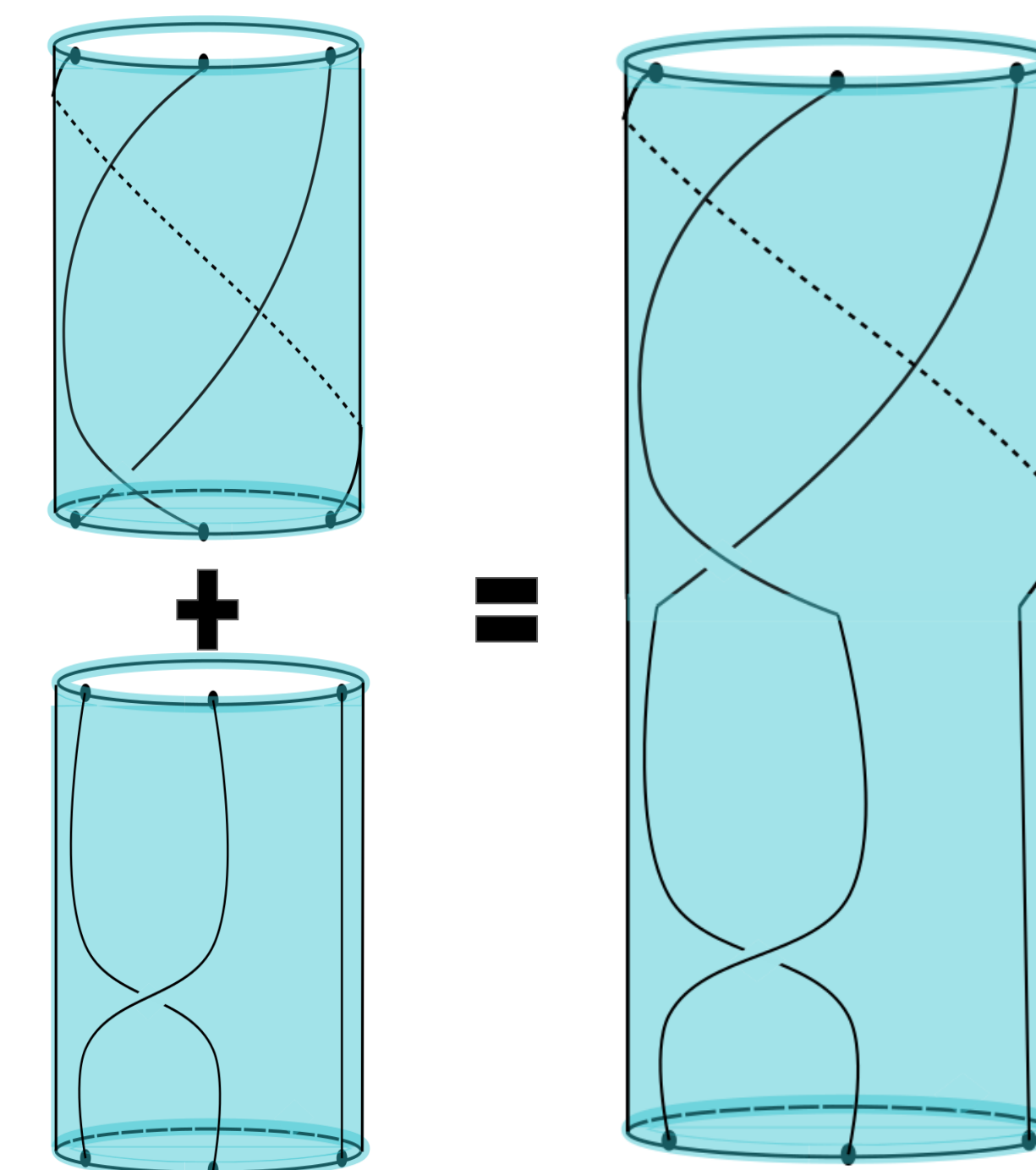
The genus of a braid is the minimal genus of its thickened surface diagram. The braid above has **genus 0**.

New subgroup of VB_n

Definition. VB_n^0 is the set of all genus-0 virtual braids.

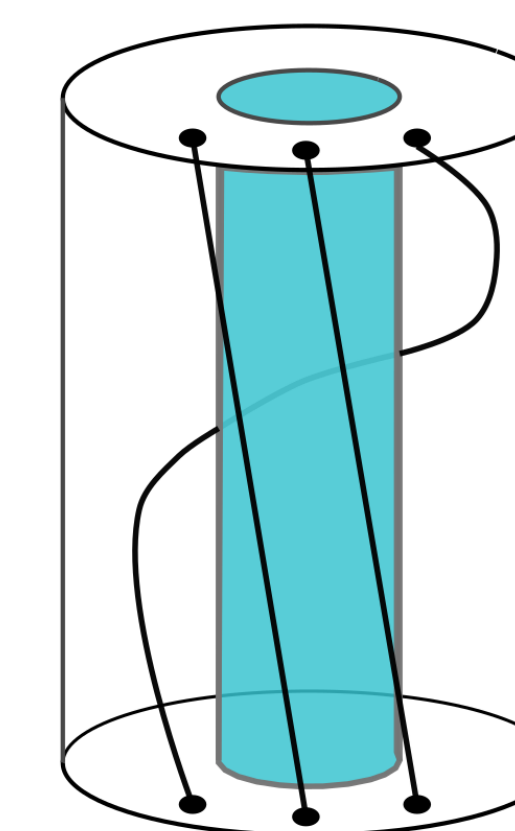
Proposition. VB_n^0 is a subgroup of the virtual braids.

Proof Sketch. Thickened braid diagrams form a group under the operation of stacking diagrams. The bijection between virtual braids and thickened surface diagrams is then an isomorphism, and the proposition follows.

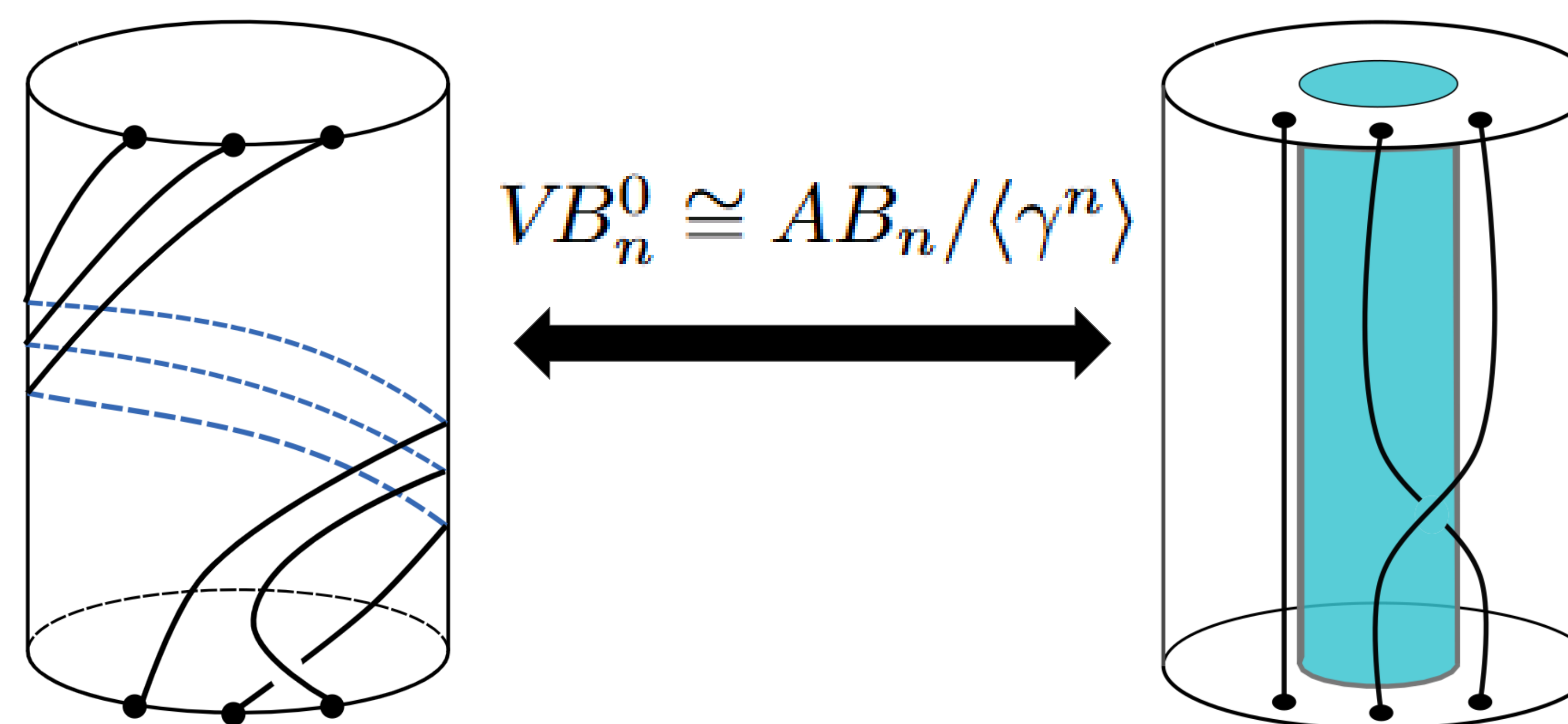


Are genus-0 braids annular braids?

$$\gamma \in AB_3$$



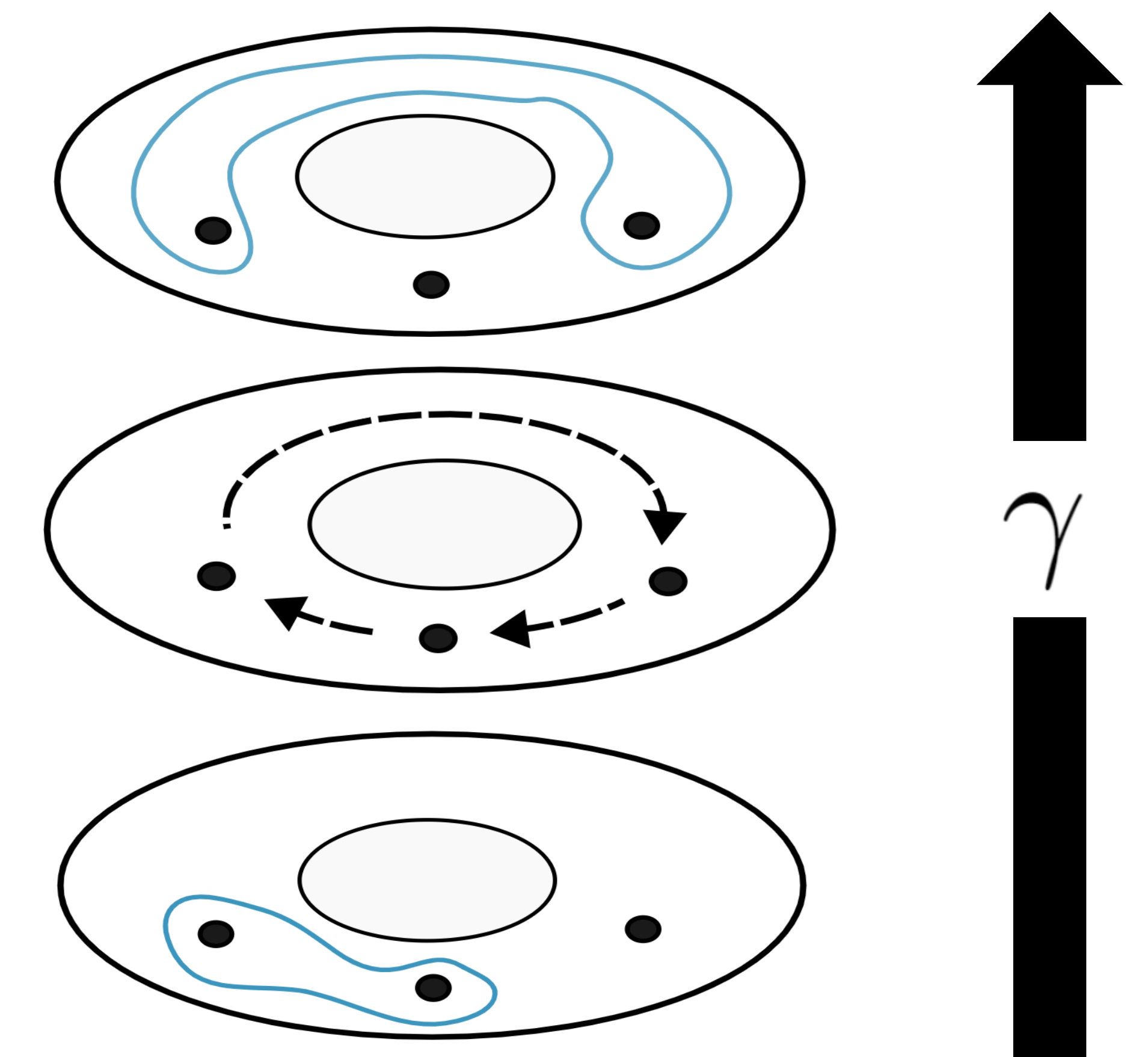
Main Theorem



Corollaries

These results follow from facts known about the annular braid group:

- VB_n^0 acts on curves on annulus
- VB_n^0 is linear
- $VB_n^0 \leq \text{Aut}(F_n)$



Future Work

- Extend action on curves for genus greater than 0.
- Compute bounds on genus of a given virtual braid.